

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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Anonymous submission

ABSTRACT

Evaluating generative models remains a fundamental challenge: automatic metrics correlate only weakly with human perception, whereas human evaluation, though more faithful, is costly and noisy because annotators differ systematically in expertise, attention, and diligence. Pairwise comparison, in which a participant selects the preferred of two items, yields markedly more consistent judgments than absolute scoring, yet recovering a global quality ranking from such comparisons demands principled aggregation. Bradley-Terry models are the standard instrument for this task; existing variants, however, either disregard heterogeneity in rater reliability or depend on optimization procedures that carry no convergence guarantees, which in turn limits their robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry model that explicitly represents rater quality and, through an Expectation-Maximization algorithm with closed-form updates, down-weights or removes unreliable participants while guaranteeing monotonic improvement of the observed-data likelihood. Across eight language and image evaluation datasets, BBQ delivers faster and more reliable convergence, well-calibrated uncertainty estimates, and consistently more robust and interpretable rankings than competing Bradley-Terry baselines, with its advantage most pronounced precisely under the noisy, crowdsourced conditions in which rater reliability is hardest to guarantee. BBQ thereby establishes a practical foundation for more reliable and cost-effective human evaluation of generative models.

1 INTRODUCTION

Evaluating generative models is a central and persistent difficulty, most acutely for large language models (LLMs) and image generators, where the metrics used in practice systematically fail to track human preference. Surrogate measures such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, capture only a narrow facet of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluation is therefore indispensable whenever a meaningful ordering between models is required, and it is for this reason that contemporary leaderboards and comparative studies of both large language models and Learned Image Compression (LIC) methods rest squarely on human preference data.

Human evaluation, however, is expensive and time-consuming, which makes the design of protocols that are simultaneously efficient and resistant to subjectivity and noise a first-order concern. Pairwise comparison, in which a participant selects between two items rather than assigning an absolute score, is an especially effective and practical instrument for this purpose (Zerman et al., 2018; Wang et al., 2023): it imposes a lighter cognitive burden on participants and elicits substantially more consistent judgments. Exhaustive collection of comparisons is nonetheless infeasible, since the number of distinct pairs grows quadratically in the number of items. Under a limited comparison budget, simple summary statistics such as win rates fail to yield a coherent global quality score, because any single win rate is confounded by the quality of the specific opponents against which an item happened to be compared.

Producing the global ranking that leaderboards require therefore hinges on aggregating pairwise comparisons effectively, which places a premium on aggregation methods that are at once robust and computationally efficient. The Bradley-Terry (BT) model (Bradley & Terry, 1952) meets this

need by estimating a latent skill for each item from the observed comparison outcomes, and it has become the standard instrument for aggregating pairwise judgments across a wide range of domains.

Useful as it is, the standard Bradley–Terry model exhibits two limitations that become decisive in practice. The first is algorithmic. Its parameters can be estimated by several procedures, including iterative scaling (Ford Jr, 1957), gradient descent, and the minorization-maximization (MM) algorithm (Hunter, 2004). For the basic BT model the MM algorithm is guaranteed to converge to the global maximum, whereas the gradient-based updates that are closely related to the Elo rating system, despite their prevalence in practice, offer no such guarantee (Hunter, 2004). This gap widens for the extensions that real studies actually require: once the model incorporates factors such as home-field advantage (Agresti, 2010), comparisons over more than two items (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to a local optimum (Hunter, 2004). The second limitation is statistical. Human evaluation is inherently noisy—participants rush, guess, or lose focus—so reliability varies markedly across raters, and when this variability is left unmodeled, a small number of unreliable raters can distort the estimated item scores and destabilize the resulting ranking.

We address both limitations with a single, unified model: a Bayesian Bradley–Terry variant that jointly infers item quality and rater reliability. To the best of our knowledge, this is the first method to combine the Bradley–Terry framework with a Bayesian formulation that explicitly models rater quality while admitting closed-form EM updates. Prior approaches to noisy raters rely on gradient-based optimization and therefore inherit its lack of convergence guarantees; our formulation instead yields stable, guaranteed convergence together with directly interpretable parameters. Two design choices make this possible. First, the Bayesian treatment supplies principled regularization and quantifies epistemic uncertainty in the estimated skills. Second, we introduce a rater-specific quality parameter that encodes how consistent and trustworthy each participant is and, through it, calibrates the influence each participant exerts on the aggregate ranking. From a latent-variable formulation of this model we derive an Expectation-Maximization algorithm that estimates item and rater parameters efficiently in closed form and is guaranteed to improve the observed-data likelihood monotonically. The mechanism is graded rather than binary: partially reliable raters continue to contribute genuine information while the influence of inattentive or inconsistent participants is systematically attenuated. Because the model is Bayesian and reasons explicitly about rater quality, it further returns uncertainty estimates for item scores, yielding rankings that are both more interpretable and more robustly comparable across studies.

We substantiate these claims on a diverse suite of human evaluation datasets spanning both the language and image domains, and we introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹ designed specifically to expose how aggregation methods behave under heterogeneous rater reliability. Across these settings BBQ converges faster, tolerates noisy comparisons markedly better, and produces more consistent rankings than the standard Bradley–Terry model and its existing rater-aware and Bayesian variants. The underlying problem—transforming noisy pairwise comparisons into a reliable global ranking—recurs well beyond generative-model evaluation, and the framework applies directly wherever that problem arises. Taken together, these contributions advance the practice of human evaluation toward studies of AI-generated content that are more cost-efficient, more interpretable, and more reproducible.

2 RELATED WORK

Ranking items is a recurring objective of human studies, and the elicitation format materially affects the quality of the resulting data. Asking a participant to choose between two items, rather than to assign an absolute score to a single item, sharpens sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). This advantage explains why so many machine-learning leaderboards are built on pairwise comparisons: participants register a preference between two alternatives, and these preferences are subsequently aggregated into an overall ranking. The Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025) are prominent instances, both collecting pairwise judgments and aggregating them with variants of the Bradley-Terry (BT) model.

¹The IHQ dataset will be released upon acceptance.

The Bradley-Terry model (Bradley & Terry, 1952), originally introduced to rank competitors in sports, provides a probabilistic account of the likelihood that one item is preferred over another and thereby converts a collection of pairwise comparisons into a global ranking. This combination of interpretability and generality has made it a cornerstone of studies spanning games, consumer preference, and numerous other domains, and it has attracted a family of extensions that broaden its reach. The Plackett-Luce model generalizes BT from pairwise comparisons to full rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a distribution over permutations as a product of successive BT probabilities (Luce et al., 1959). Further extensions target specific structural features of the data, modeling home-field advantage (Agresti, 2010), admitting ties (Rao & Kupper, 1967), or handling comparisons between groups rather than individuals (Huang et al., 2006).

Parameter inference for the basic BT model is most commonly carried out by maximum likelihood estimation (MLE). Zermelo (1929) proposed an iterative scheme for computing these estimates that has since been widely adopted, and Lange et al. (2000) later showed it to be a special case of the minorization-maximization (MM) family, which iteratively optimizes surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM to generalized BT models and characterized the conditions under which it converges to the MLE. The distinction that matters for the present work is one of guarantees: for the classical BT model the optimization is convex, so MM converges to the global optimum (Hunter, 2004), whereas for extensions such as home-field advantage, multiple comparisons, or ties, MM may converge only to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model have been developed to inject prior knowledge and regularization into estimation (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Of particular relevance to our approach, Caron & Doucet (2012) established that MM algorithms admit an Expectation-Maximization (EM) interpretation, which in turn enables Bayesian inference via Gibbs sampling; the resulting formulation yields tractable complete-data likelihoods and guarantees convergence of the associated Markov Chain Monte Carlo procedure. Beyond these computational benefits, the Bayesian perspective smooths posterior estimates, mitigates susceptibility to local maxima, and quantifies uncertainty in the estimated item skills.

A further and, for our purposes, critical line of work relaxes the assumption that all comparisons are equally reliable. Because participant quality varies widely in crowdsourced evaluation, several approaches model annotator reliability explicitly (Chen et al., 2013). The closest of these is that of Chen et al. (2013), who proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters: it places Gaussian priors on both the item and rater parameters and introduces scaling factors in the likelihood to modulate each annotator’s influence. Posterior inference, however, is carried out by gradient-based optimization, which is difficult in high-dimensional parameter spaces because of local optima and, more fundamentally, carries no convergence guarantee. Our EM-based method departs from this design precisely here, guaranteeing monotonic improvement of the likelihood. A simplified variant of this idea, the Elo-based rater model developed by Google Research, is widely deployed in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and in related research (Mentzer et al., 2020; Ballé et al., 2025).

Against this backdrop, our contribution occupies a previously unfilled position: to the best of our knowledge, no prior work combines the Bradley-Terry model with a Bayesian formulation that explicitly models rater quality while also admitting closed-form EM updates. Where Chen et al. (2013) rely on Gaussian priors and gradient-based optimization without convergence guarantees, we adopt conjugate priors and an EM algorithm that guarantees monotonic likelihood improvement. And where the Elo-based rater model used in CLIC (CLIC, 2025) is a simplified heuristic built on Elo-style updates, our method supplies a principled Bayesian treatment that additionally furnishes uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Recovering a reliable ranking of items from a set of noisy pairwise comparisons is a fundamental problem in machine learning and statistics, and we now formalize it. Consider a set of K items

that are repeatedly compared in pairs by a set of R raters, and let D denote the data comprising the outcomes of these comparisons. For two items i and j , Bradley & Terry (1952) model the comparison outcome as

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is the parameter associated with item $k \in \{1, 2, \dots, K\}$ and represents its latent skill rating. This formulation infers interpretable item scores directly from the observed wins and losses, but it rests on the simplifying assumption that every comparison is equally reliable. In human evaluation that assumption is routinely violated: participants differ in expertise, attention, and diligence, and these differences translate into judgments of uneven and inconsistent reliability.

To account for this heterogeneity, we augment the model with a rater-specific quality parameter $q_r \in [0, 1]$ that governs a two-component behavioral mixture: with probability q_r rater r issues an informed judgment drawn from the Bradley-Terry model, and with the complementary probability $1 - q_r$ the rater guesses at random, equivalent to a fair coin flip between the two items. The probability that rater r ranks item i above item j is therefore

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

Summing over all raters and item pairs, the log-likelihood of the observed data is

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ is the total number of comparisons of items i and j by rater r and $w_{r,ij}$ is the number of those comparisons in which rater r ranks item i above item j . This likelihood admits no closed-form maximizer, so the parameters λ and q cannot be obtained analytically. A generic iterative optimizer such as gradient descent is applicable but offers no convergence guarantee and requires careful learning-rate tuning. We instead adopt the Expectation–Maximization (EM) algorithm, which dispenses with learning-rate tuning and guarantees both a monotonic increase of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

To render the likelihood amenable to EM, we recast the Bradley-Terry model within the Thurstonian framework (Diaconis, 1988), following Caron & Doucet (2012). Under this construction a comparison between items i and j is modeled as a race in which each item is assigned a random arrival time, Y_i and Y_j , and the arrival times are taken to be exponentially distributed:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. Because the minimum of two independent exponential variables recovers exactly the ratio of rates, this construction reproduces the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

To obtain a tractable complete-data likelihood for EM, we introduce two families of latent variables. The first is a Bernoulli indicator

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which records whether the c -th comparison of items i and j by rater r follows the Bradley-Terry component of the mixture rather than the random-guessing component. Conditioning on these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times over the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent-variable representation is what renders the subsequent EM update tractable while still accounting explicitly for rater-specific quality.

3.3 EXPECTATION-MAXIMIZATION UPDATES

We use EM to compute maximum a posteriori (MAP) estimates of the model parameters λ and q , treating the per-comparison Bradley-Terry indicators and the unobserved arrival times of the Thurstonian construction as latent variables. By the standard EM guarantee, the procedure converges to a stationary point of the posterior.

We first place conjugate priors on the two parameter families. The item skills receive a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, conjugate to the exponential arrival-time model, and each rater's quality parameter receives a Beta prior, $q_r \sim B(\alpha, \beta)$, conjugate to the Bernoulli reliability indicators. These conjugate choices are what make the maximization step available in closed form.

The algorithm iteratively maximizes the expected complete-data log-posterior, conditioned on the current parameter estimates (λ^*, q^*) , alternating the following two steps until convergence.

E-step: Expectation In the E-step we form the expected complete-data log-posterior, denoted Q : the expectation of the log-posterior over all observed and latent variables, taken with respect to the latent variables conditioned on the observed data and the parameter estimates from the previous iteration. It is defined as

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood ℓ_c factorizes into three additive components, corresponding to the latent arrival times, the reliability indicators, and the observed comparison outcomes:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the parameters by maximizing Q . Substituting the expected sufficient statistics computed in the E-step reduces the original, analytically intractable optimization to a pair of simple closed-form updates. Both updates are driven by a single quantity: the posterior probability that a given comparison from rater r was generated by the Bradley-Terry component rather than by random guessing. This quantity, denoted γ , acts as the weight assigned to a rater's judgment according to how closely it aligns with the model's current predictions, and is given by

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j under the current estimates. The posterior responsibility γ thus quantifies the model's confidence that a comparison was informative rather than random, with larger values marking more trustworthy comparisons; it is exactly this soft weighting that lets the model discount unreliable judgments without discarding them outright.

The update for a rater's quality q_r takes the form of a regularized weighted average. Its numerator accumulates the rater's effective number of wins—each win weighted by the posterior responsibility

γ that the judgment was informative rather than random—augmented by the shape hyperparameter of the Beta prior, while its denominator normalizes by the rater’s total number of comparisons together with the prior parameters. The estimate consequently rises when a rater’s judgments align frequently with the model’s predictions and falls otherwise. The update is

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for an item’s skill λ_i is likewise a ratio of two interpretable quantities. Its numerator aggregates the effective wins of item i across all raters, each again weighted by the responsibility γ , and so measures the total positive evidence in favor of item i ; its denominator accumulates the comparisons in which item i participated and acts as the corresponding normalizer. Both terms are regularized by the Gamma prior hyperparameters. The update is

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R [\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)}] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

A complete derivation of the Expectation-Maximization algorithm, including the E-step expectations and the closed-form M-step solutions, is provided in Appendix B.

4 EXPERIMENTAL RESULTS

We evaluate our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against two strong and directly relevant baselines: the Bayesian Bradley-Terry model (Bayes-BT) (Caron & Doucet, 2012), which is Bayesian but rater-agnostic, and a gradient-descent-based BT model that models rater quality (Crowd-BT) (Chen et al., 2013), which is rater-aware but lacks convergence guarantees. These two baselines isolate the two ingredients BBQ unifies, so that any advantage can be attributed to their combination rather than to either alone. For Crowd-BT we use the Google Research implementation (Google, 2025), the same code employed by CLIC (2025) and Ballé et al. (2025). All hyperparameter settings are reported in Appendix C.

The evaluation spans both modalities in which human preference data are central. It covers human-preference benchmarks for large language models—HUMAINE (Team, 2025) and MT-Bench (Zheng et al., 2023)—and for image compression—WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), and ConHa (Aczel & Wattenhofer, 2024)—together with our newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To probe the effect of rater quality directly, we partition IHQ into a screened and an unscreened subset, which lets us isolate how low-quality, high-quality, and mixed comparisons each influence model behavior. The construction of IHQ is detailed in Appendix E, and Appendix D together with Table 2 summarizes every dataset in terms of comparisons, raters, and items. Collectively these datasets span several orders of magnitude in scale and rater reliability, providing a demanding testbed for aggregation methods.

Because ground-truth rankings are unavailable for real datasets, we establish reference orderings in two complementary ways. On simulated data, where a true ordering exists by construction, all three methods recover it as the number of samples grows, confirming their basic correctness. On crowdsourced data, we approximate the ground truth by the ranking obtained from the full dataset, and we verify that this approximation is reasonable by inspecting the top-ranked item: on every dataset except IHQ-unscreened, all three models agree on the top item. The exception is diagnostic rather than incidental. On IHQ-unscreened, both Crowd-BT and Bayes-BT fail to identify the reference image as the highest-quality item, whereas BBQ correctly recovers it—an early indication that explicit rater-quality modeling is decisive precisely when reliability is heterogeneous.

A reliable aggregation method should reproduce the same ordering when the study is repeated, and we quantify this stability through bootstrapping, which estimates the variability of the recovered rankings; the procedure is described in Appendix F. One caveat applies to the WD and HiFiC studies,

which selected comparison pairs actively rather than uniformly—HiFiC via a binary-search strategy and WD via maximum information gain. Because bootstrapping over such adaptively chosen pairs does not faithfully reproduce the original sampling process, the bootstrapping results on these two datasets should be interpreted with corresponding caution.

We report two complementary metrics that together characterize both the top of the ranking and its entirety. Top-1 agreement is the fraction of bootstrapped samples whose best item coincides with that of the full dataset, and is the pertinent measure when identifying the single best model is the goal. Kendall’s Tau (τ) measures the ordinal correlation between two rankings (Kendall, 1938), with higher values indicating greater stability, and captures the consistency of the ordering as a whole.

4.1 PERFORMANCE ACROSS DATASETS

We first compare the three methods across all datasets on both ranking accuracy and stability, with Top-1 agreement and Kendall’s Tau summarized in Table 1. As expected, datasets that afford more comparisons per model—MT-Bench, WD, and HiFiC—are generally the easier ones. HUMAINE is an instructive exception to this trend: despite its large comparison count, it does not become uniformly easy, which demonstrates that raw comparison volume is not the sole determinant of difficulty and that rater consistency and diversity exert an independent influence on performance.

Across the eight datasets BBQ delivers the strongest overall stability and robustness. It attains the highest Top-1 agreement on every dataset, sharing the top position with the baselines on three, and it recovers the overall ordering most accurately—the best Kendall’s Tau—on five of the eight datasets, ranking a close second on the remaining three.

Crucially, the three datasets on which BBQ places second in Kendall’s Tau are exactly those with high-quality, homogeneous rater pools, so the pattern of results is coherent rather than mixed. On MT-Bench all models recover the top item perfectly; the WD dataset of Ballé et al. (2025) comprises only five raters, likely the authors themselves and hence exceptionally careful; and in IHQ-screened the raters were explicitly filtered for quality. When rater reliability is uniformly high, there is little variability for a rater-quality model to exploit, so the advantage of BBQ over simpler models narrows precisely where one would predict—a boundary condition that the remaining experiments confirm from the opposite direction.

The IHQ experiments, which contrast the screened and unscreened subsets, make this pattern unmistakable. Whenever low-quality raters are present, as in IHQ-all and IHQ-unscreened, BBQ outperforms both Crowd-BT and Bayes-BT by a substantial margin. All three models attain their best Top-1 accuracy on the screened subset, reaffirming the value of rater selection, but they diverge sharply once screening is removed. Although Crowd-BT also models rater quality explicitly, its Top-1 accuracy falls markedly on the full dataset—from 98.57% on the screened subset to 33.15% on the unscreened one—most plausibly because crowdsourced raters contribute fewer than 40 comparisons each, leaving too little per-rater signal for reliable quality estimation under noise. BBQ, by contrast, remains robust without any screening: its Top-1 accuracy declines only slightly, from 99.80% to 61.92%, and it stays the strongest method by a wide margin, which is the central empirical claim of the paper.

This robustness matters because uniformly high rater quality is difficult and expensive to guarantee in practice: large-scale crowdsourcing inevitably introduces variability, rigorous screening procedures are costly, and subjective factors can compromise even diligent annotators. In exactly these prevalent conditions, BBQ offers a principled means of extracting useful signal from partially reliable raters while suppressing the influence of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

Table 1 already indicates that datasets with more comparisons per model tend to yield better performance. The total number of comparisons can be raised along two distinct axes—by recruiting more raters or by asking each rater to perform more comparisons—and disentangling these axes clarifies where each method’s advantage originates.

Figure 1 isolates the two axes for the three BT variants. The left column varies the number of raters while holding comparisons per rater at the maximum, and the right column varies comparisons per

Table 1: Performance of the three BT-based aggregation methods across eight datasets, comparing BBQ (ours) against the Bayes-BT and Crowd-BT baselines. Top block: Top-1 agreement [%]; bottom block: Kendall’s Tau; higher is better in both, and per-dataset best values are in bold with second-best underlined. BBQ attains the highest Top-1 agreement on every dataset and the best Kendall’s Tau on five of the eight, ranking second on the remaining three; its advantage is largest on the noisy IHQ-all and IHQ-unscreened settings, where rater reliability is heterogeneous.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	98.90	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	<u>0.9180</u>	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

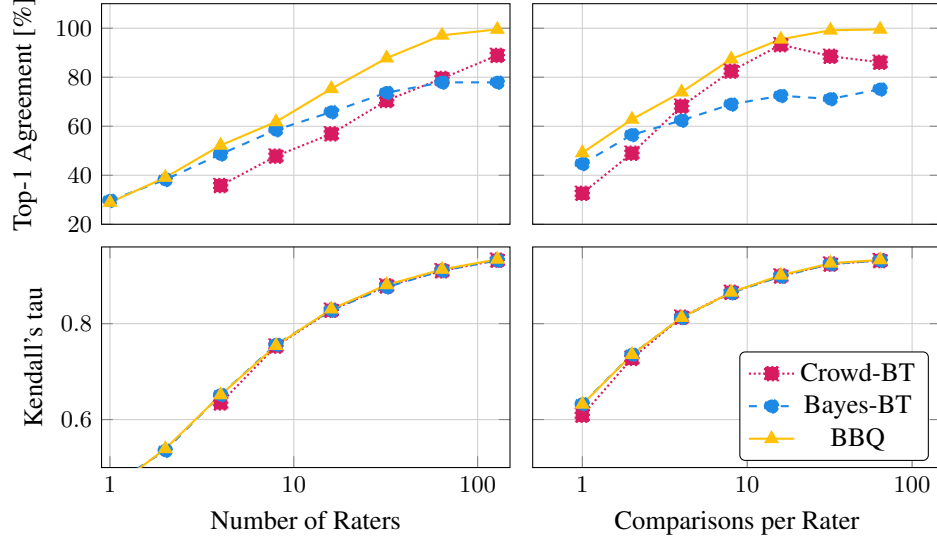


Figure 1: Scaling behavior of the three Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), and BBQ (ours)) on the IHQ-all dataset. **Left column:** performance as a function of the number of raters, with comparisons per rater held at the maximum. **Right column:** performance as a function of the number of comparisons per rater, with the number of raters held at the maximum. Both Top-1 agreement (top row) and Kendall’s τ (bottom row) improve steadily as either quantity grows. Top-1 agreement clearly separates the methods—BBQ dominating in the data-rich regime—whereas Kendall’s τ is comparable across methods. Crowd-BT fails to converge with only one or two raters, exposing the value of BBQ’s convergence-guaranteed EM updates; in the sparse regime Crowd-BT and BBQ perform comparably, while as raters or comparisons accumulate BBQ increasingly surpasses Bayes-BT.

rater while holding the number of raters at the maximum. The methods separate clearly in Top-1 accuracy, whereas Kendall’s τ remains comparable across them. Most tellingly, Crowd-BT fails to converge with only one or two raters, a failure that BBQ never exhibits and that directly reflects the convergence guarantee of EM over gradient descent.

In the sparse regime, Crowd-BT and BBQ perform comparably, which is expected: BBQ requires both multiple raters and multiple comparisons per rater before it can distinguish rater qualities reli-

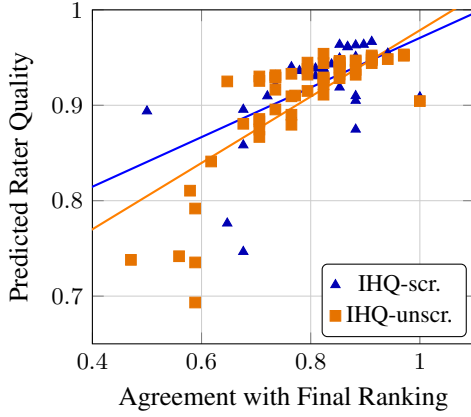


Figure 2: Predicted rater quality (y-axis) against each rater’s agreement with the final ranking (x-axis) for the IHQ datasets, with one point per rater; triangles denote the filtered (screened) subset and squares the unfiltered (unscreened) subset. In both subsets predicted quality rises with agreement, and the fitted regression lines confirm a positive association—stronger for the unfiltered subset ($r = 0.7238$) than the filtered one ($r = 0.5510$)—indicating that BBQ’s estimated rater quality tracks how closely a rater agrees with the consensus.

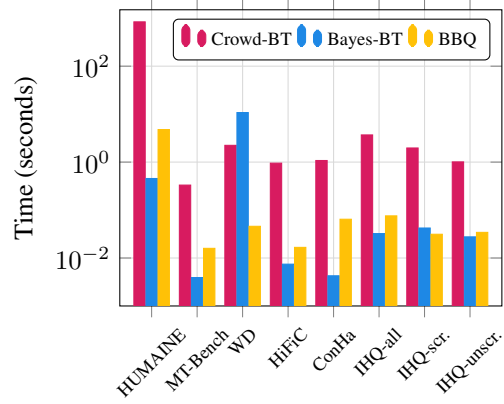


Figure 3: Average computation time in seconds (log scale) to process a single bootstrapped sample, reported for the three methods across the eight datasets. Crowd-BT incurs substantial cost on several datasets—most acutely on HUMAINE—whereas BBQ remains uniformly fast across all datasets, converging within seconds in every case.

ably. As either quantity increases, however, BBQ progressively pulls ahead of Bayes-BT, converting the additional data into a widening margin. Bayes-BT is the weakest method overall, yet it can edge out Crowd-BT when data per rater or the number of raters is very limited, because Crowd-BT cannot estimate rater quality dependably in such sparse settings. The three methods thus occupy complementary regimes, and BBQ is the only one that remains strong across the full range.

4.3 RATER QUALITY

Figure 2 examines whether BBQ’s estimated rater-quality parameter is meaningful by relating it to each rater’s agreement with the final ranking. The relationship is clearly positive: raters who agree more closely with the consensus are assigned higher quality by the model. The Pearson correlation is $r = 0.551$ over the 50 raters of the filtered dataset and a stronger $r = 0.724$ over the 62 raters of the unfiltered dataset—the greater value in the unfiltered case reflecting the wider spread of rater reliability that the parameter is designed to capture.

As anticipated, the unscreened dataset contains several raters exhibiting both low agreement and low predicted quality, and some fall below the chance level of 50% agreement, systematically disagreeing with the majority. Such raters are exactly the failure mode that motivates explicit rater-quality modeling. BBQ identifies them reliably and assigns them correspondingly low quality scores, thereby down-weighting their influence on the final ranking and neutralizing the noise they would otherwise introduce—the mechanism that underlies its robustness on the unscreened data.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average wall-clock time each method requires to process a single bootstrapped sample per dataset, a quantity that directly governs the feasibility of large-scale, repeated-resampling evaluations. BBQ converges within a few seconds on every dataset. Crowd-BT is slower than BBQ throughout, and dramatically so on datasets with many comparisons: on HUMAINE a single bootstrap iteration takes roughly 15 minutes, two to three orders of magnitude longer than BBQ. Bayes-

BT is slightly faster than BBQ on most datasets but converges slowly on WD. BBQ’s consistent speed follows directly from its closed-form EM updates, which secure rapid convergence even at scale.

This comparison in fact understates BBQ’s efficiency advantage: Crowd-BT is a highly optimized implementation written in C (Kernighan & Ritchie, 1988), whereas BBQ runs in plain NumPy without any dedicated optimization, so a compiled or vectorized implementation could only widen the gap. BBQ is therefore not merely more robust and stable but also eminently practical for large-scale human-preference studies and any workflow that relies on repeated bootstrapping. This convergence of accuracy, stability, and speed within a single method makes BBQ a compelling default for real-world deployment.

4.5 CONFIDENCE INTERVALS

Beyond point rankings, we assess how faithfully each method quantifies uncertainty, since well-calibrated intervals are what license claims of statistical significance between items. In a controlled simulation, raters compare equally skilled items, so the null hypothesis H_0 of no true difference holds by construction; each method then estimates skills and constructs uncertainty intervals used to test for a significant difference (Appendix G). The resulting Type I error rates appear in Figure 6. At the 99% confidence level the target error rate is 1%. Crowd-BT and BBQ both track this target closely, at slightly lower rates, whereas Bayes-BT is markedly conservative at roughly 0.1%. BBQ therefore yields well-calibrated uncertainty estimates on par with Crowd-BT, while Bayes-BT systematically understates the false-positive rate and so is less suitable for significance testing.

5 LIMITATIONS

The scope of BBQ’s advantage follows directly from what it models, and we state its boundaries plainly. Because BBQ exploits variation in rater reliability, its benefit over simpler models diminishes when raters are uniformly reliable; in such homogeneous datasets there is little variability for the rater-quality parameter to leverage. Guaranteeing uniformly high rater quality is itself costly and often infeasible at scale, so this limitation delineates rather than undercuts the setting BBQ targets: it is most valuable in realistic crowdsourced evaluations and least differentiated when evaluations are already carefully curated.

A second limitation concerns the supported feedback format. BBQ is presently defined for pairwise comparisons, whereas many human-evaluation studies elicit ratings, full rankings, or other multi-way responses; extending the model to accommodate these formats is a natural direction for broadening its applicability.

Methodologically, our stability estimates rest on a finite number of bootstrap resamples. This provides a practical and consistent measure of stability, but larger-scale studies would further corroborate the robustness we observe. The model also assumes independence across comparisons and does not model contextual or order effects, which can shape human judgments in practice.

Finally, although BBQ scales well, very large numbers of items or raters could still strain a plain, uncompiled implementation—a practical rather than fundamental constraint, given that its closed-form updates are directly amenable to compilation and vectorization.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Its central contribution is an Expectation-Maximization algorithm with closed-form updates that estimates item skills and individual rater quality simultaneously, down-weighting or removing the influence of unreliable participants and thereby producing more stable and accurate rankings. Unlike the gradient-based approaches it supersedes, the algorithm guarantees monotonic likelihood improvement and rapid convergence.

These properties translate into consistent empirical gains. Across eight diverse language and image datasets, BBQ achieved the highest Top-1 agreement on every dataset and the best Kendall’s Tau

540 on the majority, confirming both its robustness and its reliability. Its advantage is greatest in ex-
541 actly the setting that motivates it—noisy, crowdsourced evaluations in which raters are non-experts,
542 contribute few comparisons, and vary widely in attentiveness and diligence—which establishes that
543 modeling rater quality is decisive when evaluation reliability is heterogeneous. The Bayesian for-
544 mulation additionally supplies principled uncertainty estimates for item scores, and we showed that
545 the resulting error bars are well-calibrated and can be used to test whether differences between items
546 are statistically significant. Finally, the estimated rater-quality parameters correlate with agreement
547 to the final ranking, validating BBQ’s ability to identify reliable evaluators.

548 By uniting rater-quality modeling, guaranteed convergence, and calibrated uncertainty in one effi-
549 cient method, BBQ offers a practical, interpretable, and broadly applicable approach to aggregating
550 noisy pairwise comparisons, and thereby advances human-evaluation methodology toward studies
551 of AI-generated content that are more cost-effective, interpretable, and reproducible.

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REFERENCES

- Till Aczel and Roger Wattenhofer. Conditional hallucinations for image compression. *arXiv preprint arXiv:2410.19493*, 2024.
- Eldridge S Adams. Bayesian analysis of linear dominance hierarchies. *Animal Behaviour*, 69(5): 1191–1201, 2005.
- Alan Agresti. *Analysis of ordinal categorical data*. John Wiley & Sons, 2010.
- Jona Ballé, Luca Versari, Emilien Dupont, Hyunjik Kim, and Matthias Bauer. Good, cheap, and fast: Overfitted image compression with wasserstein distortion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 23259–23268, 2025.
- Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Francois Caron and Arnaud Doucet. Efficient bayesian inference for generalized bradley–terry models. *Journal of Computational and Graphical Statistics*, 21(1):174–196, 2012.
- Xi Chen, Paul N Bennett, Kevyn Collins-Thompson, and Eric Horvitz. Pairwise ranking aggregation in a crowdsourced setting. In *Proceedings of the sixth ACM international conference on Web search and data mining*, pp. 193–202, 2013.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael Jordan, Joseph E Gonzalez, et al. Chatbot arena: An open platform for evaluating llms by human preference. In *Forty-first International Conference on Machine Learning*, 2024.
- CLIC. Clic 2024: Challenge on learned image compression. <https://archive.compression.cc/2024/>, 2025. Accessed: 2025-09-12.
- Arthur P Dempster, Nan M Laird, and Donald B Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society: series B (methodological)*, 39(1): 1–22, 1977.
- Persi Diaconis. Group representations in probability and statistics. *Lecture notes-monograph series*, 11:i–192, 1988.
- Lester R Ford Jr. Solution of a ranking problem from binary comparisons. *The American Mathematical Monthly*, 64(8P2):28–33, 1957.
- Rafael C Gonzalez. *Digital image processing*. Pearson education india, 2009.
- Google. Google, 2025. URL <https://github.com/google-research/google-research>. Accessed: 2025-09-12.
- John Guiver and Edward Snelson. Bayesian inference for plackett-luce ranking models. In *proceedings of the 26th annual international conference on machine learning*, pp. 377–384, 2009.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Tzu-Kuo Huang, Ruby C Weng, Chih-Jen Lin, and Greg Ridgeway. Generalized bradley-terry models and multi-class probability estimates. *Journal of Machine Learning Research*, 7(1), 2006.
- David R Hunter. Mm algorithms for generalized bradley-terry models. *The annals of statistics*, 32(1):384–406, 2004.
- Frederick Jelinek. *Statistical methods for speech recognition*. MIT press, 1998.
- Maurice G Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938.
- Brian W. Kernighan and Dennis M. Ritchie. *The C Programming Language*. Prentice Hall, 1988.

648 Kenneth Lange, David R Hunter, and Ilsoon Yang. Optimization transfer using surrogate objective
649 functions. *Journal of computational and graphical statistics*, 9(1):1–20, 2000.

650

651 Mabyduck Ltd. Mabyduck, 2025. URL <https://www.mabyduck.com/>. Accessed: 2025-09-
652 12.

653 R Duncan Luce et al. *Individual choice behavior*, volume 4. Wiley New York, 1959.

654

655 Fabian Mentzer, George D Toderici, Michael Tschannen, and Eiríkur Agustsson. High-fidelity gener-
656 ative image compression. *Advances in neural information processing systems*, 33:11913–11924,
657 2020.

658 Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic
659 evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association
660 for Computational Linguistics*, pp. 311–318, 2002.

661

662 Robin L Plackett. The analysis of permutations. *Journal of the Royal Statistical Society Series C:
663 Applied Statistics*, 24(2):193–202, 1975.

664 Pejaver V Rao and Lawrence L Kupper. Ties in paired-comparison experiments: A generalization
665 of the bradley-terry model. *Journal of the American Statistical Association*, 62(317):194–204,
666 1967.

667

668 Prolific AI Team. Humaine: Human-ai interaction evaluation dataset, 2025. URL [https://
669 huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset](https://huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset).

670 Kexin Wang, Yunlong Zhao, Qianqian Dong, Tom Ko, and Mingxuan Wang. Mospc: Mos prediction
671 based on pairwise comparison. *arXiv preprint arXiv:2306.10493*, 2023.

672

673 Zhou Wang, Eero P Simoncelli, and Alan C Bovik. Multiscale structural similarity for image quality
674 assessment. In *The thirty-seventh asilomar conference on signals, systems & computers, 2003*,
675 volume 2, pp. 1398–1402. Ieee, 2003.

676 Emin Zerman, Vedad Hulusic, Giuseppe Valenzise, Rafał K Mantiuk, and Frédéric Dufaux. The
677 relation between mos and pairwise comparisons and the importance of cross-content comparisons.
678 *Electronic Imaging*, 30:1–6, 2018.

679 Ernst Zermelo. Die berechnung der turnier-ergebnisse als ein maximumproblem der wahrschein-
680 lichkeitsrechnung. *Mathematische Zeitschrift*, 29(1):436–460, 1929.

681

682 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
683 Zi Lin, Zhuohan Li, Dacheng Li, Eric. P Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
684 Judging llm-as-a-judge with mt-bench and chatbot arena, 2023.

A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items, where rater r has quality q_r : with probability q_r the rater follows the Bradley–Terry model and with probability $1 - q_r$ chooses at random. The derivation below is a standard EM (Dempster et al., 1977) argument, presented in full, that incorporates these rater-specific reliabilities into the Bayesian estimation framework and yields the closed-form updates used in the main text.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We adopt conjugate priors throughout: a Gamma prior $\lambda_i \sim \Gamma(a, b)$ on each item skill and a Beta prior $q_r \sim B(\alpha, \beta)$ on each rater quality.

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j: j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*)^{\frac{1}{2}}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) , holding fixed the expected sufficient statistics computed in the E-step. Because the priors are conjugate, each maximization admits a closed-form solution, derived below.

Update for q_r The update for each rater quality q_r follows from maximizing Q with respect to q_r , including the contribution of the Beta prior.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i follows from maximizing Q with respect to λ_i , including the contribution of the Gamma prior.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters.

The Bayes-BT and BBQ models introduce only two kinds of hyperparameters: the parameters of the prior distributions and the convergence threshold. For both models, we declare convergence when no Elo score changes by more than 1 between successive iterations. The Elo score is obtained from the skill parameter λ as

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO.SCALE.FACTOR to 400.

For the skill parameters in both Bayes-BT and BBQ we use a Gamma prior with shape 5 and rate 0.1, and for the rater-quality parameter in BBQ we use a Beta prior with $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate on a deliberately diverse collection of human-preference datasets spanning both the language and image domains, summarized in Table 2.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which contributes crowd-worker model-comparison judgments on multi-turn dialogues.

For image compression, we use three datasets that differ sharply in their rater regimes: WD (Ballé et al., 2025), a dense, expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which targets conditional generative models. Their substantial variation in scale, rater count, and rater expertise makes the collection a broad and demanding testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected in a user study on the CLIC2024 (CLIC, 2025) data conducted via the *Mabyduck* platform. It comprises two-alternative-forced-choice (2AFC) judgments over 28 generative image-compression models. To make rater quality an experimentally controllable variable, we provide two subsets: (i) a **screened** subset, restricted to raters who passed pre-screening checks for attention and display quality, and (ii) an **un-screened** subset containing all raters. The screened subset represents higher-quality raters, while the unscreened subset deliberately reproduces the noisy conditions characteristic of large-scale human evaluation.

Table 2: Summary of the datasets used in our experiments, reporting the number of comparisons, raters, and models together with the average comparisons per rater and per model. The IHQ dataset was collected by us on the Mabyduck platform in both 2AFC and 3AFC settings, and we provide screened and unscreened subsets: the screened subsets retain only raters who passed pre-screening tests for attention and display quality, ensuring higher reliability, whereas the unscreened subsets include all raters. The wide variation across rows—spanning five to nearly two thousand raters and tens to thousands of comparisons per rater—is what makes this collection a stringent test of robustness to rater heterogeneity.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

Together, these datasets let us study both large-scale, relatively clean benchmarks and smaller, noisier settings in which modeling rater reliability becomes critical—the full spectrum over which an aggregation method must remain dependable.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected on the Mabyduck platform (Ltd., 2025), shown in Figure 4. Each trial posed the question “Which image looks more similar to the reference image?”, disallowed ties, and drew the compared pair uniformly at random—the uniform sampling being precisely what makes rater-level bootstrapping on IHQ well founded, in contrast to the actively sampled WD and HiFiC studies.

To make rater quality experimentally controllable, we partition the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and additionally completed training comparisons in which they selected the higher-quality image, reinforcing the intended evaluation criteria before contributing data.

Figure 5 presents four representative pre-screening images. The first two are standard color-blindness tests in which raters must correctly identify the number embedded in each pattern, and the final two are shape-detection tests probing sensitivity to low-contrast objects—a light gray shape on a white background and a dark gray shape on a black background. Together these tests ensure that only raters with adequate visual acuity enter the screened subset, providing the controlled contrast against which the unscreened subset’s heterogeneity is measured.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT model rater quality explicitly, we bootstrap over raters rather than over individual comparisons; this preserves each rater’s full comparison distribution and therefore yields a fair assessment of stability under the same conditions the models are designed for. We draw 10,000 bootstrap resamples of raters for every dataset, with a single exception: on HUMAINE a single Crowd-BT bootstrap iteration takes roughly 15 minutes (see Section 4.4), so we use 1,000 resamples there to keep computation tractable.

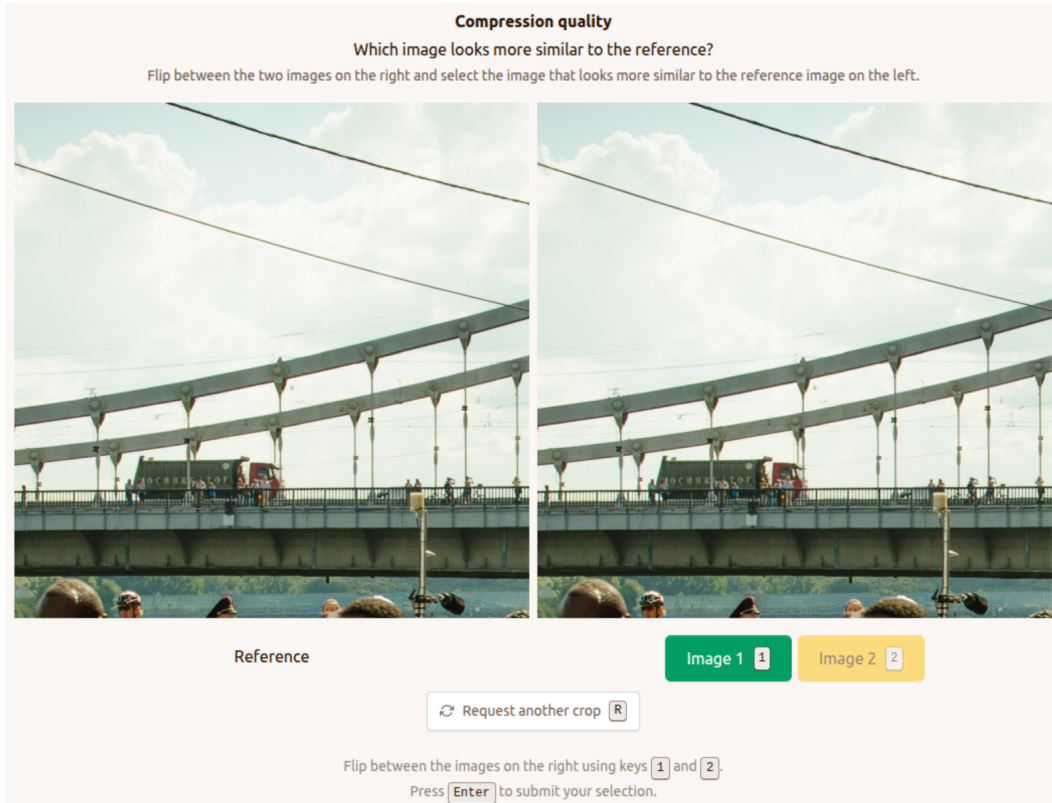


Figure 4: The Mabyduck user-study platform used to collect the pairwise comparisons. The reference image is displayed on the left, and the rater selects between two compressed candidates shown on the right.

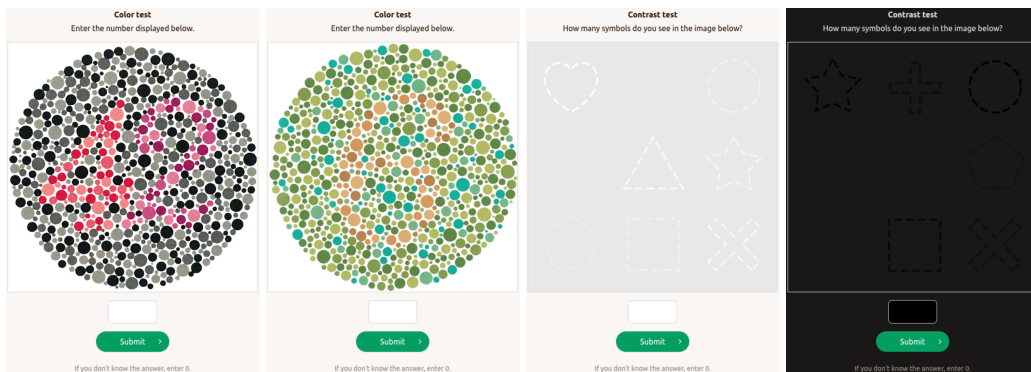


Figure 5: Four representative images from the rater pre-screening process. The first two are color-blindness tests in which raters identify the number embedded in each pattern; the last two are shape-detection tests probing sensitivity to low-contrast objects—one light gray shape on a white background and one dark gray shape on a black background.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing a ranking, quantifying uncertainty is essential for judging whether an observed difference between items is statistically significant, and the three methods do so by fundamentally different mechanisms.

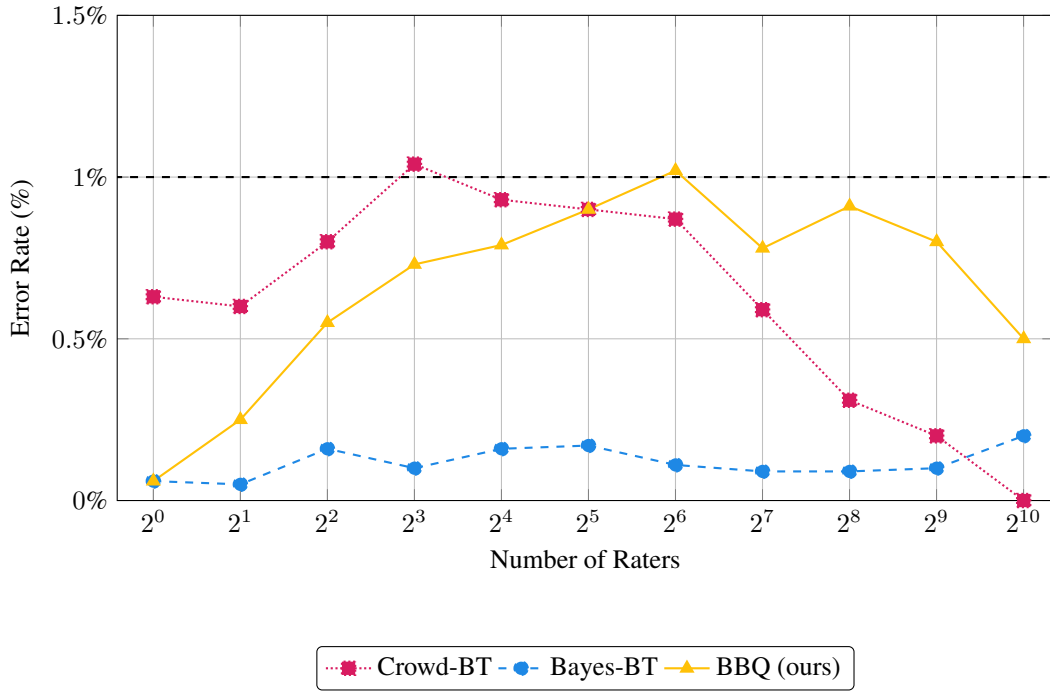


Figure 6: Type I error rate as a function of the number of raters, with the dashed line marking the expected 1% rate at the 99% confidence level. Crowd-BT and BBQ remain close to this target, whereas Bayes-BT is consistently and substantially below it.

The Bayesian methods—Bayes-BT and BBQ—quantify uncertainty directly through the posterior distribution over item skills: each skill carries a Gamma prior that is updated by the observed pairwise comparisons, and the resulting posterior credible intervals are mapped onto the Elo scale to make them comparable across items.

Crowd-BT, being non-Bayesian, instead relies on a frequentist approximation built from a second-order Taylor expansion of the log-likelihood around its maximum. Inverting the Hessian yields the covariance matrix, whose diagonal entries are the per-item variances; these are converted to 99% confidence intervals via

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to the 99% confidence level. A sharply peaked likelihood yields narrow intervals, signaling high certainty, whereas a flat likelihood yields wide intervals, signaling greater uncertainty.

To evaluate the fidelity of these uncertainty estimates, we design a controlled simulation in which the correct answer is known by construction. Each trial pits two items of exactly equal skill (a 50–50 win probability), and we generate the pairwise comparison data by fair coin flips. For each trial we simulate multiple raters and apply every method to estimate Elo scores together with their associated uncertainty.

Formally, the null hypothesis H_0 asserts that the two items are equally strong. For each trial we compute the 99% confidence (or credible) interval of each item’s skill estimate, and a method is deemed to falsely reject H_0 whenever these intervals fail to overlap—declaring a significant difference where none exists. The metric of interest is thus the observed Type I error rate: the frequency with which a method wrongly concludes that the items differ.

We run 10,000 trials with 50 comparisons per rater, varying the number of raters to trace how calibration scales with data volume, and we compare each empirical error rate against the theoretical target of 1% at the 99% confidence level. As Figure 6 shows, Crowd-BT and BBQ hold close to the expected 1%, confirming that their uncertainty estimates are well-calibrated. Bayes-BT, by contrast,

1188 is systematically over-conservative, producing error rates well below the target and thereby revealing
1189 that its credible intervals are poorly calibrated for significance testing.
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